

Domain Information Groper (도메인 정보 탐색기)

1. 역할

: DNS 네임서버에 질의(Query)를 보내고 응답을 받아 분석하는 도구.

2. 비교

- nslookup: 대화형 모드 지원, 윈도우 사용자에게 친숙. 하지만 결과가 요약되어 나오거나 OS에 따라 결과가 다를 수 있음.
- dig: ****Raw Data(날것의 데이터)****를 보여줌. DNS 헤더 정보, 응답 시간, 쿼리 옵션 등을 정밀하게 조작 가능. 해커나 보안 전문가들이 선호.

3. 명령어 형태

```
dig @[DNS서버IP] [도메인] [레코드타입] [옵션]
```

[명령어]

목적	명령어 예시
IP 조회	dig ubiedu.co.kr
네임서버 조회	dig ubiedu.co.kr ns
메일 서버 조회	dig ubiedu.co.kr mx
결과만 심플하게	dig ubiedu.co.kr +short
특정 서버 지정	dig @8.8.8.8 ubiedu.co.kr
경로 추적	dig ubiedu.co.kr +trace
존 전송(공격)	dig @[NS서버] [도메인] axfr

4. 실습

실습환경 : 칼리 리눅스 > cmd 터미널

피해서버 : zonetransfer.me or 본인의 존파일에 서브도메인이 여러개 작성되어있는 서버를 켜서
준비 !

1) 기본조회 (A레코드)

```
dig hrpark.com
```

: ANSER SECTION에 IP주소가 나오는지 확인

```
(rim@kali)-[~]
$ dig hrpark.com

; <<>> DiG 9.20.11-4+b1-Debian <<>> hrpark.com
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 62959
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: d0de93ec9cc9931701000000692563866a8ca6c594362304 (good)
;; QUESTION SECTION:
;hrpark.com.                IN      A

;; ANSWER SECTION:
hrpark.com.                 86400   IN      A      172.16.11.100

;; Query time: 0 msec
;; SERVER: 172.16.11.100#53(172.16.11.100) (UDP)
;; WHEN: Tue Nov 25 03:06:30 EST 2025
;; MSG SIZE rcvd: 83
```

[HEADER]

status: NOERROR -> 정상 응답.

status: NXDOMAIN -> 도메인
없음 (오타 났거나 존재하지 않음).

[QUESTION SECTION]: 내가
무엇을 물어봤는지.

[ANSWER SECTION] 실제 정답
(IP 주소 등)

Stats (맨 아래): Query time (응답
속도), SERVER (누가
응답했는지)

2) 특정 정보 수집(MX, NS)

```
dig zonetransfer.me mx +short  
dig zonetransfer.me ns
```

: 공격을 수행하려면 이 도메인을 관리하는 진짜 주인(NS)를 알아야 하므로 dig로 NS레코드 조회 결과값을 보고 타겟으로 삼겠다.

[dig zonetransfer.me mx +short]

```
(rim@kali)-[~]  
$ dig zonetransfer.me mx +short  
\0 ASPMX.L.GOOGLE.COM.  
20 ASPMX3.GOOGLEMAIL.COM.  
20 ASPMX4.GOOGLEMAIL.COM.  
10 ALT2.ASPMX.L.GOOGLE.COM.  
10 ALT1.ASPMX.L.GOOGLE.COM.  
20 ASPMX2.GOOGLEMAIL.COM.  
20 ASPMX5.GOOGLEMAIL.COM.
```

[dig hrpark.com ns]

```
(rim@kali)-[~]  
$ dig hrpark.com ns  
  
; <<>> DiG 9.20.11-4+b1-Debian <<>> hrpark.com ns  
;; global options: +cmd  
;; Got answer:  
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 14043  
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 2  
  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags:;; udp: 1232  
; COOKIE: ff0f2b0d2e304e950100000069253269df0695a5fefccae6 (good)  
;; QUESTION SECTION:  
;hrpark.com. IN NS  
  
;; ANSWER SECTION:  
hrpark.com. 86400 IN NS ns.hrpark.com.  
  
;; ADDITIONAL SECTION:  
ns.hrpark.com. 86400 IN A 172.16.11.100
```

3) DNS 경로 추적

```
dig hrpark.com +trace
```

: Root(.) -> TLD(.me) -> Authoritative(zonetransfer.me) 순서로 찾아가는 계층 구조를 눈으로 확인할 수 있습니다.

a) Root Server(.)에게 물어봄

```
(rim@kali)-[~]
$ dig hrpark.com +trace

; <<>> DiG 9.20.11-4+b1-Debian <<>> hrpark.com +trace
;; global options: +cmd
.                508532  IN      NS      m.root-servers.net.
.                508532  IN      NS      b.root-servers.net.
.                508532  IN      NS      j.root-servers.net.
.                508532  IN      NS      g.root-servers.net.
.                508532  IN      NS      l.root-servers.net.
.                508532  IN      NS      d.root-servers.net.
.                508532  IN      NS      h.root-servers.net.
.                508532  IN      NS      c.root-servers.net.
.                508532  IN      NS      a.root-servers.net.
.                508532  IN      NS      k.root-servers.net.
.                508532  IN      NS      e.root-servers.net.
.                508532  IN      NS      f.root-servers.net.
.                508532  IN      NS      i.root-servers.net.
.                508532  IN      RRSIG   NS 8 0 518400 20251207170000
20251124160000 61809 . Hs449vTb18CcGOvpY55A2MQ7xW35YdKvJ0hhX2jdon4ZJAioxpORoR
```

b) TLD Server(.me)를 찾음

```
com.          172800 IN      NS      a.gtld-servers.net.
com.          172800 IN      NS      b.gtld-servers.net.
com.          172800 IN      NS      c.gtld-servers.net.
com.          172800 IN      NS      d.gtld-servers.net.
com.          172800 IN      NS      e.gtld-servers.net.
com.          172800 IN      NS      f.gtld-servers.net.
com.          172800 IN      NS      g.gtld-servers.net.
com.          172800 IN      NS      h.gtld-servers.net.
com.          172800 IN      NS      i.gtld-servers.net.
com.          172800 IN      NS      j.gtld-servers.net.
com.          172800 IN      NS      k.gtld-servers.net.
com.          172800 IN      NS      l.gtld-servers.net.
com.          172800 IN      NS      m.gtld-servers.net.
com.          86400  IN      DS      19718 13 2 8ACBB0CD28F41250A8
0A491389424D341522D946B0DA0C0291F2D3D7 71D7805A
```

c) Authoritative Server 찾음

```
hrpark.com.   172800 IN      NS      ns1.afternic.com.
hrpark.com.   172800 IN      NS      ns2.afternic.com.
```

d) 최종 IP 찾음

```
hrpark.com.   3600  IN      A      76.223.54.146
hrpark.com.   3600  IN      A      13.248.169.48
;; Received 71 bytes from 173.201.66.69#53(ns2.afternic.com) in 136 ms
```

=> 특정 아이피로 찾고싶으면 이렇게 입력해서 찾을것

`dig @172.16.11.100 hrpark.com`

4) Zone transfer 공격 (AXFR)

: 앞서 찾은 네임서버 에게 직접 데이터 요청

```
dig axfr @ns.hrpark.com hrpark.com
```

: 구글 DNS 말고 너(관리서버)에게 직접 물어보겠다. 존파일 전체를 전송해달라!!!

```
(rim@kali)-[~]
$ dig axfr @ns.hrpark.com hrpark.com

; <<>> DiG 9.20.11-4+b1-Debian <<>> axfr @ns.hrpark.com hrpark.com
; (1 server found)
;; global options: +cmd
hrpark.com.      86400    IN       SOA      ns.hrpark.com. rname.invalid.
0 86400 3600 604800 10800
hrpark.com.      86400    IN       NS       ns.hrpark.com.
hrpark.com.      86400    IN       A        172.16.11.100
hrpark.com.      86400    IN       AAAA     ::1
ftp.hrpark.com.  86400    IN       A        172.16.11.100
jl.hrpark.com.   86400    IN       A        172.16.11.100
ns.hrpark.com.   86400    IN       A        172.16.11.100
wp.hrpark.com.   86400    IN       A        172.16.11.100
ww1.hrpark.com.  86400    IN       A        172.16.11.100
ww2.hrpark.com.  86400    IN       A        172.16.11.100
ww3.hrpark.com.  86400    IN       A        172.16.11.100
hrpark.com.      86400    IN       SOA      ns.hrpark.com. rname.invalid.
0 86400 3600 604800 10800
;; Query time: 0 msec
;; SERVER: 172.16.11.100#53(ns.hrpark.com) (TCP)
;; WHEN: Mon Nov 24 23:25:50 EST 2025
;; XFR size: 12 records (messages 1, bytes 347)
```

[(번외) nslookup 으로 dig 해보기]

1) Kali Linux 에서 아래 와 같이 명령어 입력

```
nslookup -type=ns zonetransfer.me
```

> 결과

```
(rim@kali)-[~]  
$ nslookup -type=ns zonetransfer.me  
Server:      8.8.8.8  
Address:     8.8.8.8#53  
  
Non-authoritative answer:  
zonetransfer.me nameserver = nsztm1.digi.ninja.  
zonetransfer.me nameserver = nsztm2.digi.ninja.  
  
Authoritative answers can be found from:
```


2) nsztml.digi.ninja 서버에게 zonetransfer.me 도메인의 전체 데이터를 달라고 요청

```
dig @nsztml.digi.ninja zonetransfer.me axfr
```

> 결과

```
(rim@kali)-[~]
$ dig axfr @ns.hrpark.com hrpark.com

; <<>> DiG 9.20.11-4+b1-Debian <<>> axfr @ns.hrpark.com hrpark.com
; (1 server found)
;; global options: +cmd
hrpark.com.      86400    IN       SOA      ns.hrpark.com. rname.invalid.
0 86400 3600 604800 10800
hrpark.com.      86400    IN       NS       ns.hrpark.com.
hrpark.com.      86400    IN       A        172.16.11.100
hrpark.com.      86400    IN       AAAA     ::1
ftp.hrpark.com.  86400    IN       A        172.16.11.100
jl.hrpark.com.   86400    IN       A        172.16.11.100
ns.hrpark.com.   86400    IN       A        172.16.11.100
wp.hrpark.com.   86400    IN       A        172.16.11.100
ww1.hrpark.com.  86400    IN       A        172.16.11.100
ww2.hrpark.com.  86400    IN       A        172.16.11.100
ww3.hrpark.com.  86400    IN       A        172.16.11.100
hrpark.com.      86400    IN       SOA      ns.hrpark.com. rname.invalid.
0 86400 3600 604800 10800
;; Query time: 0 msec
;; SERVER: 172.16.11.100#53(ns.hrpark.com) (TCP)
;; WHEN: Mon Nov 24 23:25:50 EST 2025
;; XFR size: 12 records (messages 1, bytes 347)
```

+ 대응 방안

: DNS 설정 파일(예: `named.conf`)에서 `allow-transfer` 지시자를 사용하여 신뢰된 보조 네임서버의 **IP**만 허용하도록 설정해야 합니다.

[로그확인]

- RHEL 계열 (Rocky, CentOS, Fedora):

`/var/log/messages`

(시스템의 모든 로그가 모이는 곳. DNS 로그도 여기 섞여 있음)

또는 `/var/named/data/named.run` (별도 설정 시)

```
Nov 25 12:44:13 localhost named[44165]: network unreachable resolving 'www.notion.so/HTTPS/IN': 2a06:98c1:50::ac40:2069#53
Nov 25 12:44:13 localhost named[44165]: network unreachable resolving 'www.notion.so/A/IN': 2a06:98c1:50::ac40:2069#53
Nov 25 12:45:23 localhost named[44165]: client @0x7f607c045ad8 172.16.11.192#37171 (hrpark.com): transfer of 'hrpark.com/IN': AXFR started (serial 0)
Nov 25 12:45:23 localhost named[44165]: client @0x7f607c045ad8 172.16.11.192#37171 (hrpark.com): transfer of 'hrpark.com/IN': AXFR ended: 1 messages, 12 records, 308 bytes, 0.001 secs (308000 bytes/sec) (serial 0)
Nov 25 12:45:33 localhost named[44165]: network unreachable resolving 'consumer.cloud.gist.build/HTTPS/IN': 2001:4860:02:36::6d#53
Nov 25 12:45:33 localhost named[44165]: network unreachable resolving 'consumer.cloud.gist.build/A/IN': 2001:4860:02:36::6d#53
```

2979,1 63%

- Debian 계열 (Ubuntu, Kali):

`/var/log/syslog`

(RHEL의 messages와 같은 역할)

```
2025-11-25T12:35:00.669341+00:00 utunto10 systemd[1]: Started packagekit.service - PackageKit Daemon.  
2025-11-25T12:35:38.496944+00:00 utunto10 named[861]: client @0x7ae1d38a5400 172.16.11.192#50353 (hrpark.com): zone  
transfer 'hrpark.com/AXFR/IN' denied  
2025-11-25T12:35:55.169877+00:00 utunto10 named[861]: client @0x7ae1d38a5400 172.16.11.192#52367 (hrpark.com): zone  
transfer 'hrpark.com/AXFR/IN' denied  
1598,1 Bot
```

syslog가 없다면 ? 설치를 해서 보면된다. !!!!!

```
sudo apt install rsyslog  
tail -f /var/log/syslog
```